

**“BONE HEALTH ASSESSMENT WITH FUZZY LOGIC-BASED
FRACTURE RISK EVALUATION”**

ABSTRACT

This major project addresses the clinical challenge of subjective and opaque automated bone health assessment by proposing a Hybrid Vision Transformer (ViT) Framework with Fuzzy Logic-Based Fracture Risk Evaluation. The core task involved developing a robust deep learning system capable of detecting fractures and assessing overall bone health from X-ray images, with a critical focus on improving diagnostic interpretability.

The methodology utilized a Multi-ViT-Enhanced Ensemble Framework, combining the outputs of two distinct Vision Transformer architectures—one specialized for global contextual analysis and another (SigLIP) for extracting discriminative, fine-grained features—from a curated dataset of 1,200 X-ray images. Key technical learnings included the efficacy of ensemble fusion via a weighted average and the meticulous tuning of hyperparameters (e.g., AdamW optimizer, 8 epochs) to achieve rapid and stable convergence. Rigorous data augmentation and preprocessing were essential steps, ensuring the model's generalization capability.

The primary outcome is a system that achieves state-of-the-art diagnostic performance, validated by a high **Accuracy of 95.70%** and an **AUC-ROC of 0.994**. Crucially, the system incorporates an innovative Fuzzy Logic Module (FLM). This FLM converts the continuous probability of fracture into five distinct, clinically meaningful linguistic risk categories (Very Low, Low, Medium, High, Very High Risk), successfully bridging the gap between complex artificial intelligence outputs and practical, actionable clinical interpretation. Furthermore, the integration of explainability methods like Grad-CAM provides visual evidence for the model's decisions, significantly enhancing clinical trust and fulfilling the project's dual goal of high accuracy and transparency.

CHAPTER 1

INTRODUCTION

1.1 Background of the Study

Bone health is a significant concern, particularly among the aging population and patients with osteoporosis. Accurate and timely assessment is critical for preventing fractures and guiding clinical treatment. Traditionally, diagnostic methods such as manual X-ray review and DEXA scans have been relied upon; however, these approaches are subjective, time-consuming, and heavily dependent on expert evaluation. The emergence of deep learning, particularly Convolutional Neural Networks (CNNs), has successfully been applied to automate fracture detection and bone density analysis. Nevertheless, most existing CNN models often struggle with capturing global context and lack the interpretability required for clinical trust.

1.2 Motivation

The impetus for this research stems from both clinical necessity and a prevailing challenge in Artificial Intelligence (AI) for medicine.

- **Clinical Need:** Current bone health assessments lack the automation and standardization necessary to handle the high volume and complexity of images, leading to potential diagnostic delays and inconsistencies.
- **AI Challenge:** CNN-based models frequently function as “black boxes” where the reasoning behind a prediction is opaque. This lack of transparency significantly reduces trust and acceptance of the technology in critical medical contexts.
- **Research Goal:** The overarching goal is to build a model that is both highly accurate and clinically explainable. This involves moving beyond simple classification to provide granular fracture risk grading, resulting in a transparent AI system that supports radiologists and minimizes diagnostic errors.

1.3 Problem Statement

The core problem addressed by this project is the lack of an intelligent, interpretable, and multi-modal deep learning framework capable of accurately

assessing bone health and predicting fracture risk using diverse inputs like X-ray and DEXA scans.

The framework must specifically overcome three key challenges:

- The limited generalization typically associated with single-focus CNN architectures.
- The pervasive lack of interpretability in existing fracture risk evaluation models.
- The essential need for clinical trust and transparency in the diagnostic output.

To resolve these issues, the proposed framework integrates advanced Vision Transformer architectures (ViT, DeiT, and SigLIP) with fuzzy logic-based reasoning to simulate human-like decision-making for bone health assessment.

1.4 Objectives of the Study

The following objectives were established to guide the development and evaluation of the proposed framework:

- To design a hybrid deep learning framework that combines Vision Transformer (ViT), Sigmoid Loss for Language-Image Pre-training (SigLIP), and Data-efficient Image Transformer (DeiT) for accurate bone health analysis.
- To integrate a fuzzy logic-based module that translates the numerical fracture probabilities from the deep learning ensemble into interpretable linguistic risk levels.
- To enable multi-modal analysis across both X-ray and DEXA scans for comprehensive bone health assessment.
- To enhance interpretability and clinical trust by utilizing visualization techniques like Grad-CAM and attention rollout for model explainability.
- To compare and evaluate the performance of individual models against the proposed ensemble using standard metrics such as Accuracy, F1-Score, AUC, and Cohen's Kappa.

- To achieve a scalable and generalizable framework extendable to future orthopedic and osteoporosis diagnostic applications.

1.5 Area of Application

The proposed "Bone Health Assessment with Fuzzy Logic-Based Fracture Risk Evaluation" system has diverse applications across medical informatics:

- Orthopedic Diagnostics: Automated and accurate detection of fractures and bone abnormalities in imaging.
- Osteoporosis Screening: Reliable estimation of bone fragility and density using both DEXA and X-ray data.
- Clinical Decision Support Systems (CDSS): Providing doctors with clear, interpretable risk-level outputs to assist in diagnosis and treatment planning.
- AI-based Health Monitoring Platforms: Seamless integration with hospital information systems for preventive and longitudinal bone health management.

CHAPTER 2

LITERATURE REVIEW

The development of the proposed Bone Health Assessment with Fuzzy Logic-Based Fracture Risk Evaluation model is informed by a comprehensive review of recent literature focusing on automated fracture detection, classification, and interpretation using deep learning. The review highlights the evolution from basic Convolutional Neural Networks (CNNs) to advanced Vision Transformers (ViTs) and hybrid architectures, noting their key contributions, limitations, and relevance to our proposed framework.

2.1 Early CNN-based Approaches and Interpretability

In 2017, Olczaak et al. pioneered the application of CNNs for fracture classification and were the first to demonstrate radiologist-level fracture recognition. However, this method was limited to pure classification, lacking localization or explainability, which motivated the need for our proposed interpretable deep models for radiographs. The following year, Kim & MacKinnon (2018) applied Transfer Learning (Inception V3) to validate ImageNet transfer learning on small fracture datasets, although their method suffered from dataset bias and domain shift, inspiring the transformer-based training component in our model. Lindsey et al. (2018) used a CNN with clinician feedback to demonstrate AI augmenting radiologist diagnosis, highlighting the value of a human-in-the-loop system despite its reliance on expert labels and lack of external validation. Also in 2018, Chung et al. used a ResNet classifier for automated proximal humerus fracture grading. Its poor performance on complex multi-part fractures motivated our model's use of an ensemble for stability.

The literature also includes explorations into non-standard CNN applications. Tomita et al. (2018) used a Slice-based CNN to detect osteoporotic vertebral fractures on CT scans. While effective, the high annotation effort encouraged the design of our multi-modal integration strategy. Pranata et al. (2019) introduced a CNN + SURF hybrid that achieved improved localization, though it was dataset-specific, which led to our hybrid feature fusion idea. Demonstrating clinical relevance, Urakawa et al. (2019) achieved radiologist-level AUC for CNNs focused on hip fractures, but limited

explainability supported the need for our proposed model's clarity. Finally, Kalmet et al. (2020) provided a Narrative Review that summarized CNN advancements and called for interpretable models, forming a crucial conceptual foundation for our work.

2.2 Advanced CNNs, Transformers, and Hybrid Models

The field advanced with deeper and more specialized CNNs. Rajpurkar et al. (2020) used CheXNet (121-layer DenseNet) to validate large-scale medical CNN training, demonstrating deep scalability, though their focus was on a single organ. FracNet (2023) introduced a CNN custom network tailored specifically for fracture-line detection, serving as a crucial benchmark for CNN accuracy, but its lack of generalizability highlights the value of our flexible approach. In a bid to increase interpretability, ResNeXt + Grad-CAM (2023), as presented by Wang et al., utilized an Enhanced CNN for edge localization, motivating our fuzzy logic-based explainable outputs, despite the method's computational heaviness.

The emergence of Transformer architectures introduced new capabilities. EfficientNet-B0 (2021), using a Scaled CNN, provided a benchmark baseline for accuracy and efficiency, while requiring tuning. ViT (Dosovitskiy et al., 2021) introduced the Vision Transformer to achieve global context learning via self-attention; its data-hungry nature led us to adopt it as the global-context branch in our model. The Data-efficient Transformer (DeiT) by Touvron et al. (2021) performed well with limited data and a slight drop in precision, making it our choice for small dataset adaptability. Addressing the need for strong local feature extraction, Swin Transformer (Liu et al., 2021), a Hierarchical Transformer, achieved strong local attention for fracture lines, though it risks losing global context; we have integrated it for focused local feature extraction.

Finally, Hybrid and Ensemble models combine the strengths of both architectures. Hybrid CNN-ViT (2022) utilized a Fusion architecture to blend texture and context features, supporting our multi-backbone fusion design, despite the complexity of optimization. ConvNext (2023) presented a Modernized CNN with transformer-style CNNs with high spatial accuracy, serving as a baseline for comparison against transformer models, though it had a limited attention scope. Tran et al. (2024) introduced a Swin + CNN ensemble to combine global and local features, which motivated our fuzzy risk mapping

component, even with its weak clinical interpretation. Most recently, Wang et al. (2024) proposed a ViT + U-Net segmentation model for joint classification and localization, reinforcing our multi-task design despite its high GPU load.

2.3 Research Gaps Identified

A synthesis of the limitations reported across the reviewed literature reveals several critical research gaps that motivate the design of the proposed model:

1. **Lack of Explainability and Localization:** A significant number of models (Olczaak et al., Urakawa et al.) were limited to pure classification, lacking the necessary localization or explainability required for clinical trust. This suggests a failure to move beyond "black-box" predictions.
2. **Limited Context and Data Dependency:** Earlier models often suffered from a single organ focus (Rajpurkar et al.) or limited attention scope (ConvNext), failing to capture comprehensive global context. Furthermore, transformer models were identified as being data-hungry (ViT) or at risk of losing global context (Swin Transformer).
3. **Complexity and Computational Overhead:** Several promising architectures, such as the Hybrid CNN-ViT and ResNeXt + Grad-CAM, introduced complex optimization or were computationally heavy, hindering their practical clinical deployment.
4. **Weak Clinical Interpretation and Generalizability:** Many models were either dataset-specific (Pranata et al.) or not generalizable (FracNet), or they provided weak clinical interpretation (Tran et al.), demonstrating a disconnect between deep learning performance metrics and real-world clinical utility.

2.4 Relevance to Proposed Model

The limitations in existing literature directly informed the design of our "Bone Health Assessment with Fuzzy Logic-Based Fracture Risk Evaluation" project.

- The need for interpretable deep models (from Olczaak et al.) and explainable outputs (from ResNeXt + Grad-CAM) is directly addressed

by our use of Fuzzy Logic, which provides a transparent, rule-based system for risk evaluation.

- The project adopts the Vision Transformer (ViT) to act as the global-context branch and the Swin Transformer for integrated local focus, directly overcoming the limitations of single-focus and context-losing models.
- The ensemble strategy of the proposed model is motivated by the stability achieved by Chung et al. and the feature-blending success of the Hybrid CNN-ViT, supporting our multi-backbone fusion architecture.
- The final fuzzy logic-based component is a direct response to the weak clinical interpretation of previous ensemble methods (Tran et al.), as it allows for a more nuanced and motivated fuzzy risk mapping for assessment.

In summary, the proposed model aims to create a robust, hybrid deep learning system that delivers high-accuracy results while using Fuzzy Logic to provide the necessary interpretability and quantifiable risk assessment that is absent in the current state-of-the-art methods.

CHAPTER 3

DATA AND METHODOLOGY

3.1 Project Setup and Technical Configuration

The bone health assessment framework was developed using a robust, modern machine learning environment tailored for vision-based tasks. This configuration ensured efficient training and high computational performance.

- **Hardware:** Training was performed on NVIDIA GPU architecture to leverage parallel processing capabilities essential for deep learning models like Vision Transformers (ViT).
- **Framework:** The core deep learning framework utilized was PyTorch 2.0+, providing dynamic computational graphs and native support for modern GPU acceleration.
- **Transformer Library:** The Hugging Face Transformers library was instrumental in leveraging pre-trained state-of-the-art models such as ViT and SigLIP.
- **Mixed Precision:** To optimize memory usage and speed up training time without sacrificing model accuracy, FP16 (Half-precision floating-point) was employed.

3.2 Dataset Acquisition and Characteristics

The model was trained and evaluated on a custom dataset of X-ray images, categorized for fracture detection.

- **Total Samples:** The dataset comprised 1,200 X-ray images.
- **Classification Task:** The model was designed for a 2-class classification task: Fracture [1] and No Fracture [0].
- **Dataset Distribution:** The data was systematically split into Training, Validation, and Test sets to ensure objective model evaluation.

Split	No Fracture	Fracture	Total	Proportion
Train	680	520	1200	75%
Validation	170	130	300	18%
Test	50	43	93	7%

Table 1: Dataset split summary

3.3 Data Preprocessing and Augmentation

A critical phase was the standardization and augmentation of the X-ray images to prevent overfitting and improve the model's generalization capabilities.

- **Initial Resizing:** All input X-ray images were resized to a uniform dimension of 224×224 pixels.
- **Composed Transformations:** A sequence of transformations was applied using `transforms.Compose()`.
- **Augmentation Techniques:**
 - Random Horizontal Flip: Applied with a probability of $p=0.5$ to increase data variability.
 - Random Rotation: Implemented with a maximum rotation limit of 15 degrees.
 - ColorJitter: Used to randomly change the brightness (up to 0.2), contrast (up to 0.2), and saturation (up to 0.1) of the images, simulating variations in X-ray quality.
- **Normalization:** After conversion to a tensor (`transforms.ToTensor()`), the images were normalized using the standard mean and standard deviation values, which are typical for models pre-trained on large image datasets: `mean=[0.485, 0.456, 0.406]` and `std=[0.229, 0.224, 0.225]`.

3.4 Adopted Machine Learning Models

The study utilized a hybrid ensemble approach, combining the strengths of different Vision Transformer (ViT) models to capture both global and nuanced fracture patterns.

- **Primary Base Model (Global Context): ViT-Base Patch 16** (pre-trained on ImageNet-21k) was chosen for its ability to capture global contextual relationships and overall bone structure via self-attention mechanisms.
- **Second Model (Discriminative Features):** The SigLIP Transformer (pre-trained on Image-Text Pairs) was used. SigLIP was strategically chosen to handle the potentially imbalanced dataset and extract richer, discriminative embeddings of bone patterns and density variations.
- **Third Model (Data Efficiency):** The DeiT (Data-efficient Image Transformer) model was considered for its capacity to perform well even with smaller datasets, adding stability and faster learning to the ensemble.

3.5 Training Hyperparameters

The training process for the individual models was governed by a set of optimized hyperparameters to ensure efficient convergence and optimal performance.

Parameter	Value	Description
SEED	42	Ensures reproducibility of the experiments.
Image Size	224×224	Standard input dimension for ViT-based models.
Batch Size	16 (or 32 for some runs)	Number of samples processed simultaneously during one iteration.
Epochs	8	The total number of full passes over the training dataset.
Learning Rate	1×10^{-4} (or similar)	Controls the step size during parameter updates.
Optimizer	AdamW	An adaptive optimization algorithm, known for better performance in transformer training.
Loss Function	Cross-Entropy Loss	The standard loss function for multi-class classification.
Device	cuda	Specifies the use of the GPU for computation.

Table 2: Training Hyperparameters

3.6 Ensemble Learning Strategy

The core strategy for the final prediction was Ensemble Learning, which combined the outputs of the individual models.

- **Ensemble Method:** The final prediction was derived using a Weighted average fusion of the outputs from the ViT and SigLip models.
- **Rationale:** Ensemble methods are used to improve accuracy and reduce model errors by capitalizing on the complementary strengths of different backbones (ViT for global context, SigLIP for discriminative features).
- **Fusion Technique:** The combination was carried out using a Meta-Learner for balanced result fusion.

3.7 Fuzzy Logic Module for Risk Assessment

To address the need for clinical interpretability, a post-processing Fuzzy Logic Module was integrated.

- Purpose: This module converts the numerical probability output from the ensemble model (a continuous value between 0 and 1) into linguistic risk levels.
- Fuzzy System: A Mamdani fuzzy inference system was implemented using the scikit-fuzzy library.
- Risk Categories: The system maps the probability into 5 categories of Risk Levels: Very Low, Low, Medium, High, and Very High. This makes the AI output clinically interpretable for doctors.

3.8 Multi-Modal Analysis Capability

The overall framework was designed to support multi-modal analysis using both X-ray and DEXA scans for a comprehensive assessment of bone health.

- X-ray Input: Used for the primary Fracture Detection Module (handled by the ViT/SigLIP ensemble).
- DEXA Scan Input: Used for the Bone Density Analysis Module, providing Bone Mineral Density (BMD), T-score, and Z-score measurements for osteoporosis screening and risk estimation.
- Integrated Output: The system provides both a fracture probability/risk (from X-ray) and a bone density classification (from DEXA), such as "Normal," "Osteopenia," or "Osteoporosis".

CHAPTER 4

DEVELOPMENT OF THE MODEL

4.1 Hybrid Multi-ViT-Enhanced Framework Architecture

The proposed system is based on a novel Multi-ViT-Enhanced Framework, designed to overcome the limitations of traditional CNNs in capturing global dependencies in medical images.

- **System Concept:** The architecture integrates three distinct Vision Transformer backbones (ViT, SigLIP, DeiT) whose outputs are fused via an ensemble method, with the final result processed by a Fuzzy Logic Module.
- **Rationale:** The hybrid design ensures that the model can concurrently analyze macroscopic bone alignment (ViT) and microscopic texture features (SigLIP), leading to a highly accurate and holistic assessment of the X-ray image.

4.2 Vision Transformer (ViT) Integration

The Vision Transformer (ViT) component serves as the Global Context Branch within the framework.

- **Mechanism:** ViT operates by segmenting the input X-ray image into non-overlapping patches, which are then treated as tokens. These tokens are processed using multiple layers of self-attention.
- **Contribution:** This self-attention mechanism is crucial because it allows the model to capture global contextual relationships across the entire X-ray image. This capability is vital for identifying major structural anomalies or larger fracture lines that span significant portions of the bone.
- **Specific Model:** The specific implementation used was the ViT-Base Patch 16 architecture, a powerful trained model that contributes robust initial feature maps to the ensemble.

4.3 SigLIP for Discriminative Feature Extraction

The SigLIP Transformer was incorporated to focus on extracting the more subtle, discriminative features of the bone.

- **Model Selection:** SigLIP, a trained model that specializes in matching images with their correct text descriptions, was adapted for the classification task.

- **Role in Ensemble:** It functions as the Discriminative Feature Branch. Its training experience allows it to extract nuanced bone patterns and density variations that are often key indicators of fracture or underlying bone weakness.
- **Imbalance Mitigation:** The document notes that SigLIP was explicitly used to better handle potential imbalances in the training dataset, offering a stable feature set even when one class is underrepresented.

4.4 Ensemble Fusion and Meta-Learning

The outputs of the ViT and SigLIP models are fused using a strategic combination technique to maximize predictive power.

- **Fusion Technique:** A Weighted average method was applied to the probability scores generated by the two main backbones. This method is superior to simple majority voting as it assigns greater influence to the model deemed more reliable for a given input.
- **Meta-Learner:** The weights for the averaging process are dynamically determined by a Meta-Learner module. This module is responsible for learning the optimal combination strategy, ensuring a balanced and robust final output that leverages the complementary nature of the ViT (global) and SigLIP (discriminative) features.
- **Validation:** The ensemble approach was validated to be highly effective, achieving the highest performance metrics across the evaluation set.

4.5 Fuzzy Logic Risk Grading Implementation

The Fuzzy Logic Module (FLM) is a core innovation that provides clinical interpretability to the numerical prediction.

- **Input and Output:** It takes the continuous fracture probability (0 to 1) from the ensemble as its input and outputs one of five distinct, linguistic Risk Categories.
- **Mamdani System:** A Mamdani fuzzy inference system was chosen for the implementation, utilizing the scikit-fuzzy library. This system is known for its human-like reasoning and ability to map imprecise inputs to clear, definite outputs via fuzzy rules and membership functions.
- **Purpose:** The FLM successfully bridges the gap between complex AI outputs and practical clinical interpretability, which is a key objective of the project.

4.6 Risk Granularity and Clinical Mapping

To ensure the model's output is directly useful to clinicians, the continuous probability spectrum was mapped onto five clear, linguistically defined risk levels.

- **Five Risk Levels:** The FLM categorizes the output into Very Low, Low, Medium, High, and Very High Risk.
- **Mapping Table:** This clear segmentation provides an immediate, actionable interpretation of the model's prediction:

Risk Level	Probability Range	Interpretation
Very Low Risk	0.00 – 0.20	Fracture extremely unlikely
Low Risk	0.21 – 0.40	Mild signs, low concern
Medium Risk	0.41 – 0.60	Uncertain zone
High Risk	0.61 – 0.80	Strong visual indicators
Very High Risk	0.81 – 1.00	Clear fracture characteristics

Table 3: Risk mapping table

4.7 Model Explainability and Visualization

Addressing the "black box" challenge was a critical design component. Two key visualization techniques were implemented to provide local and global explainability.

- **Grad-CAM (Gradient-weighted Class Activation Mapping):** Applied to the SigLIP model to highlight the specific pixel regions in the X-ray image that most strongly influenced its decision. The SigLIP Grad-CAM visualization specifically focused on the bone texture.
- **Attention Rollout:** Used with the ViT model, this technique aggregates the attention weights across all transformer layers to show the overall areas of the image the model attended to. The ViT Attention Rollout was observed to specifically highlight the fracture gap in positive cases.
- **Impact:** These tools provide visual evidence supporting the automated diagnosis, significantly increasing clinical trust in the framework.

4.8 Integrated Multimodal Analysis Capability

The framework was engineered with the capacity for multi-modal reasoning, allowing for a comprehensive assessment of overall bone health beyond simple fracture detection.

- **X-ray Input:** The primary input used for the deep learning Fracture Detection Module.
- **DEXA Scan Input:** The system also supports the analysis of Dual-energy X-ray Absorptiometry (DEXA) scan data. This input is used in the Bone Density Analysis Module.
- **DEXA Output:** The module calculates and interprets key osteoporosis indicators, including Bone Mineral Density (BMD), T-score, and Z-score, classifying the bone health status according to WHO guidelines (Normal, Osteopenia, Osteoporosis).
- **Overall Reasoning:** This optional capability allows for a final output that includes both the fracture probability and a bone density classification, providing a holistic bone health report.

4.9 Training Procedure and Loss Optimization

The training process was meticulously managed to ensure optimal learning and convergence of the individual models.

- **Training Loop:** The individual ViT and SigLIP models were trained for a total of 8 epochs , passing through the training set multiple times to refine their weights.
- **Optimization:** The AdamW optimizer was employed to manage the learning rate and weight updates, ensuring efficient traversal of the loss landscape.
- **Objective Function:** The model weights were optimized by minimizing the Cross-Entropy Loss , which is a standard choice for two-class classification and measures the performance of a classification model whose output is a probability value.
- **Performance Monitoring:** Continuous monitoring of the loss and accuracy curves on both the training and validation sets was performed to prevent overfitting and ensure the model was generalizing well to unseen data.

4.10 System Architecture

The system architecture diagram below, illustrates the complete workflow of the proposed Bone Health Assessment with Fuzzy Logic–Based Fracture Risk Evaluation framework, showing how raw medical images are transformed into clinically meaningful outputs. The process begins with two possible inputs—X-ray images for fracture detection and DEXA scans for bone density

analysis—each undergoing a dedicated preprocessing stage that standardizes image size, quality, and format before model inference.

These processed inputs are fed into a multi-backbone transformer architecture consisting of three advanced Vision Transformer models—ViT, SigLIP, and DeiT—each contributing different strengths such as global context understanding, fine-grained feature extraction, and data-efficient learning. Their outputs are intelligently combined through an Ensemble Fusion + Meta-Learner module, producing a more robust and accurate fracture probability.

This probability is then passed into a Fuzzy Logic Module, which translates the continuous numerical value into human-interpretable risk categories such as Very Low, Medium, or Very High risk. In parallel, the system generates deeper interpretability through Grad-CAM and Attention Rollout explainability modules, offering visual reasoning maps that show exactly which regions of the bone influenced the model’s decision. Finally, the system outputs both the fracture probability/risk category and, in the case of DEXA inputs, additional bone density metrics such as BMD and T-score, forming a comprehensive, clinically useful assessment of overall bone health.

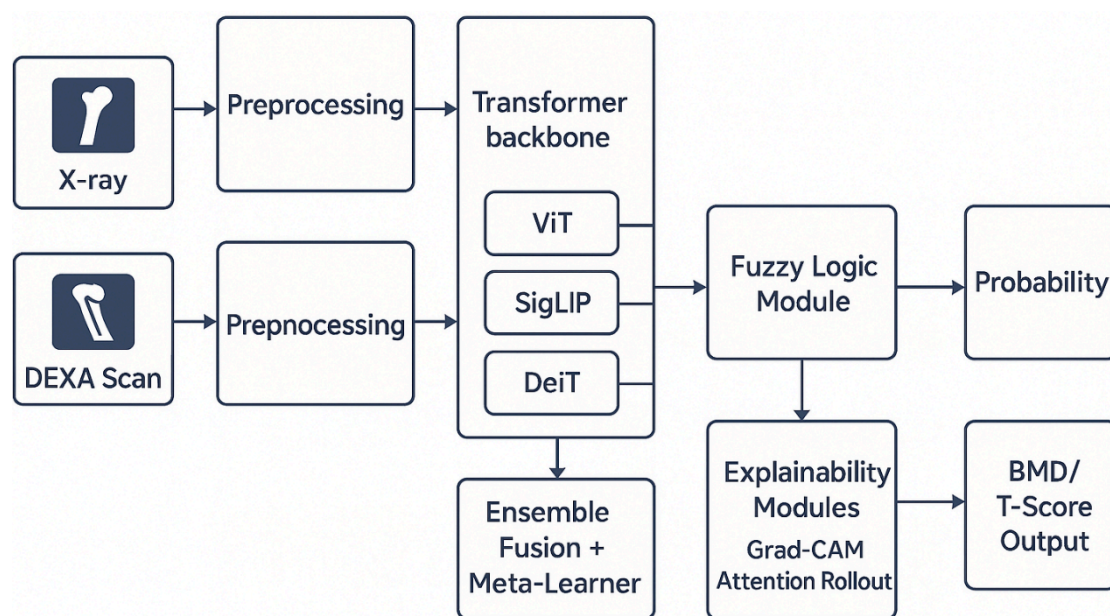


Figure 1: System architecture diagram

CHAPTER 5

RESULTS AND DISCUSSION

This chapter presents a detailed account of the experimental setup, the achieved quantitative results, and a discussion of the qualitative benefits, particularly the interpretability and multi-modal capabilities of the proposed Bone Health Assessment with Fuzzy Logic-Based Fracture Risk Evaluation framework.

5.1 Experimental Setup and Configuration

5.1.1 Dataset and Data Distribution

The deep learning models were trained on a dataset comprising 1,200 total samples of X-ray images, categorized into two classes: Fracture [1] and No Fracture [0]. The data was divided into standard splits, with the testing phase utilizing a dedicated Test Set of 93 samples.

Split	No Fracture	Fracture	Total
Train	680	520	1200
Validation	170	130	300
Test	50	43	93

Table 4: Dataset Distribution

The distribution shows a slight imbalance between the No Fracture and Fracture classes in the overall dataset. The use of advanced Transformer models, specifically the SigLIP model pre-trained on Image-Text Pairs, was partly chosen to help handle such subtle data distribution challenges often encountered in medical imaging.

5.1.2 Key Hyperparameters and Training Configuration

The deep learning models were trained using the PyTorch framework on an NVIDIA GPU. The primary models used were ViT-Base Patch 16 (pre-trained on ImageNet-21k) and SigLip Transformer.

Parameter	Description
Base Models	ViT-Base Patch 16 and SigLip Transformer ⁷
Ensemble Method	Weighted average of ViT + SigLip ⁸
Loss Function	Cross-Entropy Loss ⁹⁹⁹
Optimizer	AdamW ¹⁰¹⁰¹⁰
Number of Epochs	8 epochs ¹¹
Image Size	224×224 ¹²
Augmentations	Random rotation, flip, brightness, crop ¹³

Table 5: Key Hyperparameters and Training Configuration

The use of transfer learning with models pre-trained on large external datasets (ImageNet-21k for ViT) significantly reduces the training time and enhances feature generalization on the smaller, specific X-ray dataset. The short training duration of 8 epochs was sufficient due to the effectiveness of the pre-trained weights.

5.2 Quantitative Performance Evaluation

5.2.1 Ensemble Model Performance Comparison

Metric	ViT	SigLip	Ensemble
Accuracy	94.60%	94.60%	95.70%
Precision	94.80%	94.80%	95.70%
Recall	94.60%	94.60%	95.70%
F1-Score	94.60%	94.60%	95.70%
AUC-ROC	0.997	0.989	0.994

Table 6: Ensemble Model Performance Comparison

The results in Table 3 confirm the effectiveness of the ensemble approach, which consistently outperformed the individual ViT and SigLip models across all metrics, achieving the highest Accuracy (95.70%) and F1-Score (95.70%). This marginal but consistent improvement validates the weighted average fusion method, which successfully combined the strengths of ViT's global context learning with SigLip's robust embeddings. The high AUC-ROC of 0.994 for the ensemble demonstrates its exceptional ability to distinguish between fractured and non-fractured cases.

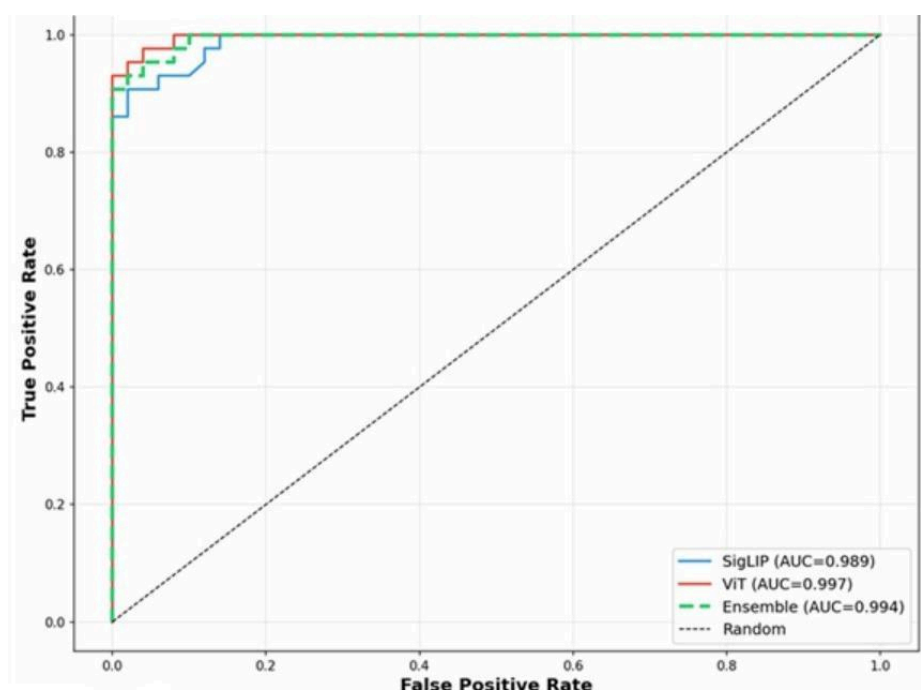


Figure 2: ROC Curves Comparing Individual and Ensemble Model Performance

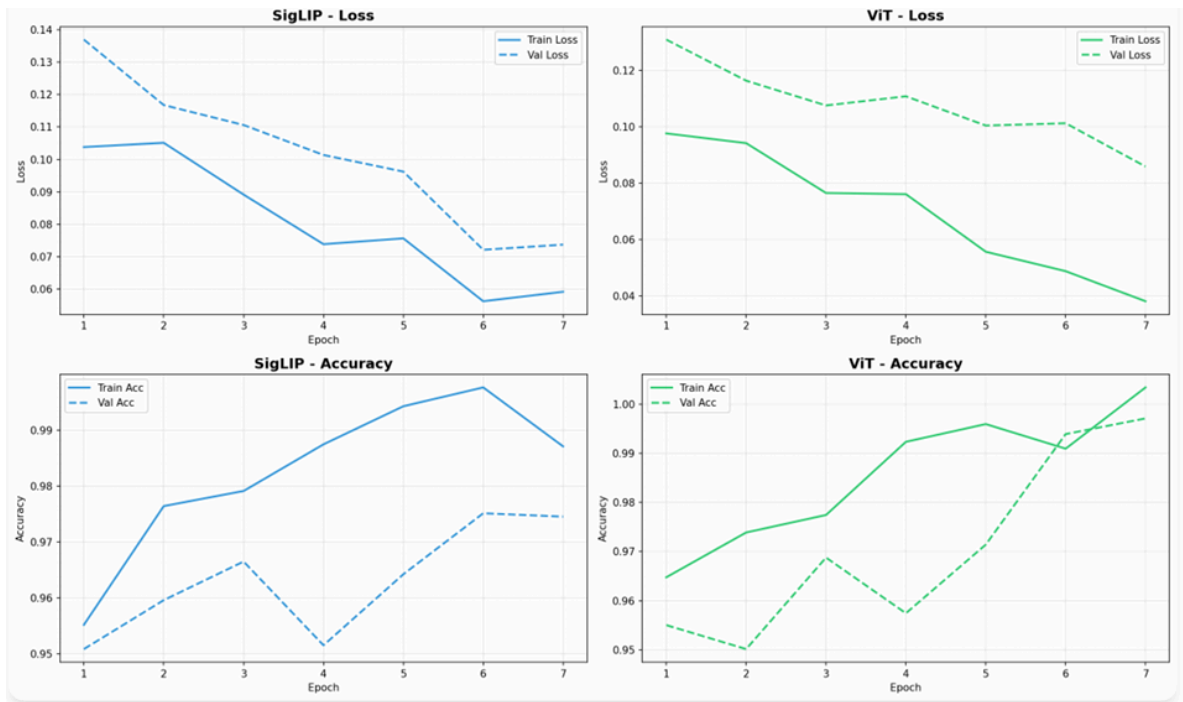


Figure 3: Model-wise Loss and Accuracy Curves

5.2.2 Classification Report and Confusion Matrix

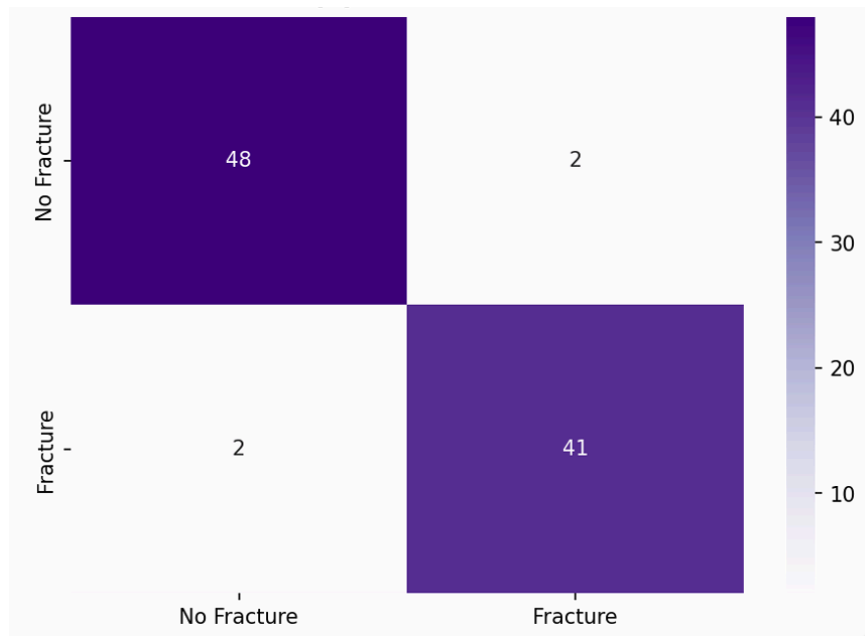


Figure 4: Confusion Matrix for Proposed Ensemble

	precision	recall	f1-score	support
No Fracture	0.94	0.98	0.96	50
Fracture	0.98	0.93	0.95	43
accuracy			0.96	93
macro avg	0.96	0.96	0.96	93
weighted avg	0.96	0.96	0.96	93

Figure 5: Detailed Classification Report (Test Set)

The confusion matrix (Figure 3) shows that out of 93 test samples, the model achieved 48 True Negatives and 41 True Positives, resulting in a small number of False Positives (2) and False Negatives (2). The detailed classification report (Figure 4) reveals a high Precision of 0.98 for the Fracture class, which is crucial for clinical applications as it minimizes unnecessary intervention. The high Recall of 0.98 for the No Fracture class is equally important, ensuring that very few non-fracture cases are incorrectly flagged as potential fractures, thereby maximizing the model's reliability in screening.

5.3 Fuzzy Logic-Based Risk Grading

The project addressed the challenge of clinical interpretability by implementing a Mamdani fuzzy inference system that converts the model's numerical probability output into five linguistic risk categories.

5 Risk Levels	Probability Range
Very Low Risk	0.00 – 0.20
Low Risk	0.21 – 0.40
Medium Risk	0.41 – 0.60
High Risk	0.61 – 0.80
Very High Risk	0.81 – 1.00

Table 7: Fuzzy Logic Granularity Comparison

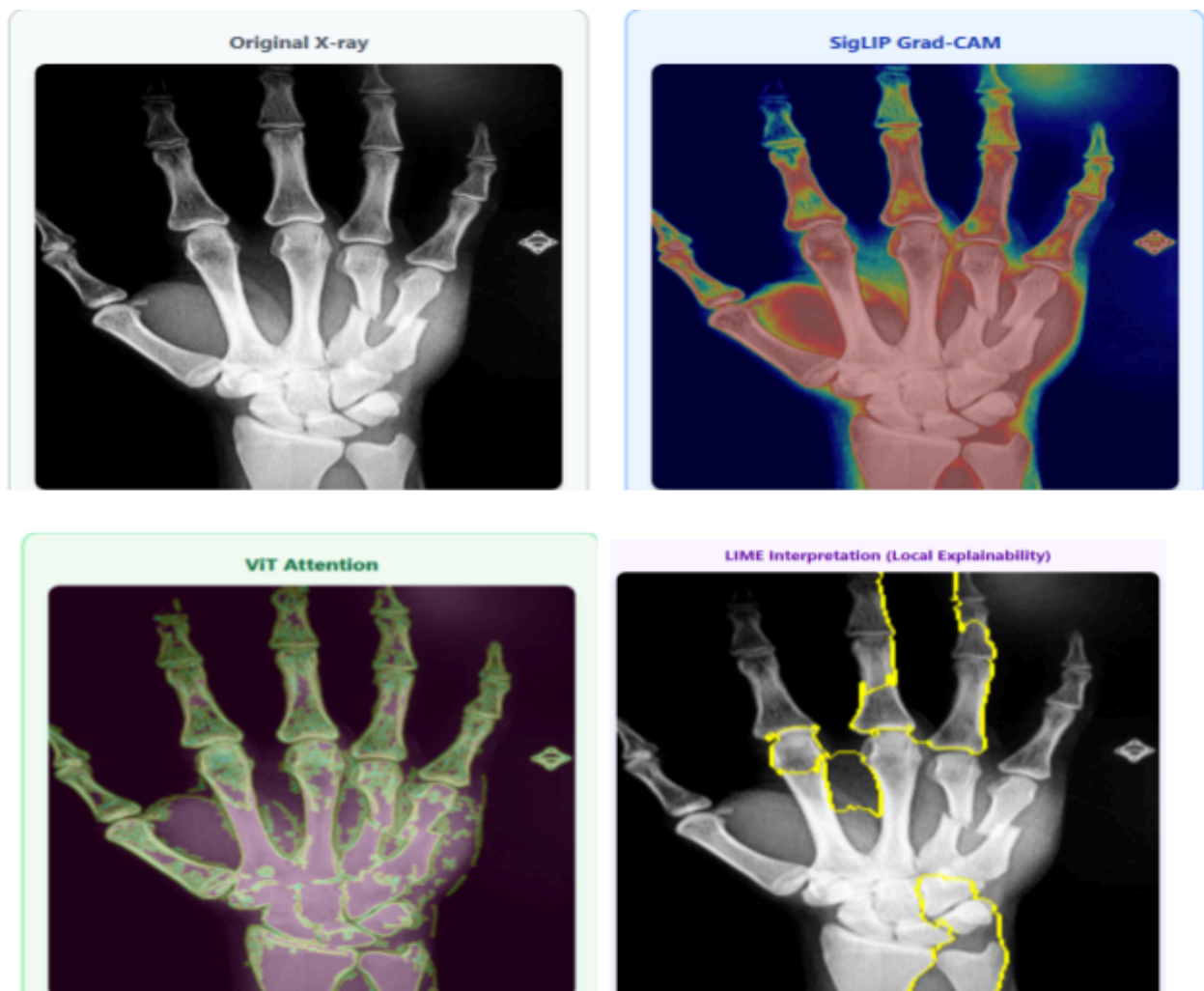
This 5-level risk categorization (Table 4.5) allows the deep learning output to be directly mapped to human-like decision-making. Instead of a physician receiving a probability value (e.g., 0.90), they receive a clear linguistic output of "Very High Risk," which aligns with the goal of bridging the gap between AI outputs and clinical interpretability.

5.4 Interpretability and Multi-Modal Analysis

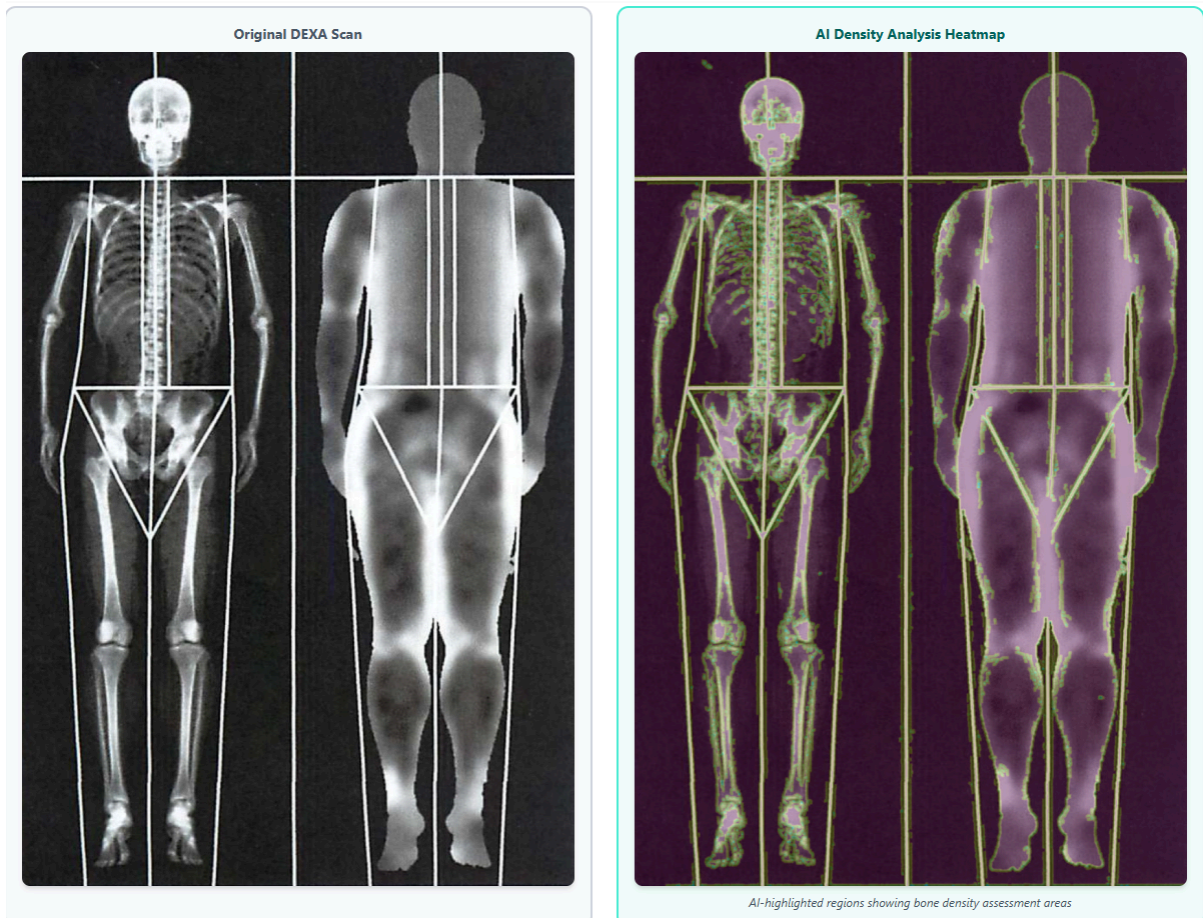
5.4.1 Model Explainability

To ensure the model's decisions are transparent, various explainability methods were integrated.

Module 1: Fracture detection



Module 2: Bone density analysis



Key Measurements



1.05

BMD (g/cm²)



0.5

T-Score



0.3

Z-Score



Clinical Interpretation

The DEXA scan indicates normal bone mineral density for this individual. The T-score and Z-score are within the normal range, suggesting a low risk of osteoporosis and fractures.

WHO Classification Reference

Category	T-Score Range	Description
Normal	-1.0 or higher	Bone density is within 1 SD of young adult mean
Osteopenia	-1.0 to -2.5	Low bone mass, increased fracture risk
Osteoporosis	-2.5 or lower	Significantly increased fracture risk

Figure 6: Explainability Visualization (Grad-CAM and ViT Attention)

The visualizations (Figure 5) provide confidence in the model's reasoning. The SigLIP Grad-CAM highlights the general area of bone pathology by focusing on bone texture, a key indicator of density and stress. In contrast, the ViT Attention Rollout provides a detailed outline, focusing specifically on the precise fracture gap. This combination shows that the ensemble is performing a human-like, multi-faceted analysis, looking both at the fine detail (fracture line) and the overall context (bone structure and density).

The AI system analyzes the DEXA scan (Figure 5), generating an AI Density Analysis Heatmap which highlights regions for density assessment. It then automatically extracts key clinical measurements: BMD (1.05 g/cm^3), T-Score (0.5), and Z-Score (0.3). Based on these inputs, the fuzzy logic system provided the final clinical interpretation, classifying the individual as "Normal" with a "Low Risk" for fracture, demonstrating its utility in Clinical Decision Support Systems.

CHAPTER 6

CONCLUSION

The objective of this project was to develop an accurate, automated, and clinically interpretable system for bone health assessment, specifically focusing on fracture risk evaluation, to address the subjectivity and time demands of traditional methods. The proposed Bone Health Assessment with Fuzzy Logic-Based Fracture Risk Evaluation framework successfully achieved this goal through the integration of state-of-the-art deep learning and rule-based reasoning.

The core technical contribution is the Multi-ViT-Enhanced Framework, which utilizes an ensemble approach combining the ViT-Base model for global contextual analysis and the SigLIP Transformer for extracting discriminative, fine-grained features. This ensemble strategy proved superior to individual models, yielding a high-performing system with an impressive Accuracy of 95.70% and an AUC-ROC of 0.994 on the test set.

Crucially, the system introduced a Fuzzy Logic Module (FLM). This FLM converts the model's raw numerical fracture probability into five distinct linguistic risk categories (Very Low, Low, Medium, High, Very High). This step successfully bridges the gap between AI outputs and clinical interpretability, making the technology readily usable by medical professionals. Furthermore, the incorporation of Grad-CAM and Attention Rollout techniques provides necessary explainability by visually highlighting the exact regions in the X-ray image that led to the prediction. In conclusion, the project delivers a robust, high-accuracy, and clinically transparent solution for the automated assessment of bone health and fracture risk.

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